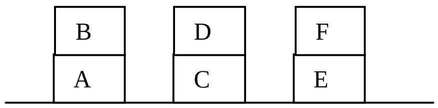
CSE4107

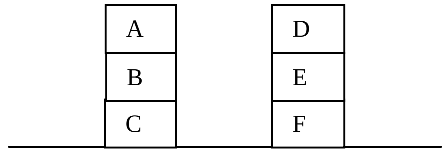
Class Test 3 Spring 2019

Marks: 10 Time: 25 min.

1. Illustrate the concept of players of a two-player board game. (3)
2. Why game playing problems are considered adversarial search problems? (2)
3. What is a partial-order plan? (1)
4. Generate an optimal plan through Forward State Space search using PDDL for the planning problem in the Blocks world presented below. Omit the predicate ‘Block’, and give priority to gain in time over gain in actions. (4)




Initial State




Goal State

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1. Explain the concept of action schemas using an example from Air Cargo Transportation problem. (3)
2. How a planner exerts intelligence through decomposition? (2)
3. What is singular expansion? (1)
4. Construct a pruned game tree using Alpha-Beta pruning. Take the sequence, [5, 3, 2, 4, 1, 3, 6, 2, 8, 7, 5, 1, 3, 4] of MINIMAX values for the nodes at the cutoff depth of 4 plies. Assume that branching factor is 2, MIN makes the first move, and nodes are generated from *left to right.* (4)

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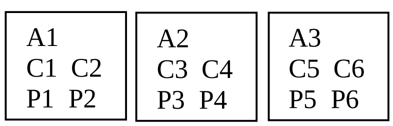
1. Explain with an example the major steps of a MINIMAX search algorithm. (3)
2. How imperfect information plays a role in game playing? (2)
3. What is an Add list? (1)
4. Generate an optimal plan using PDDL and Forward State Space search for the Air Cargo Transportation problem given below. Omit predicates ‘Airport’, ‘Cargo’ and ‘Plane’, and consider generally accepted rules and notations for such a problem. (4)

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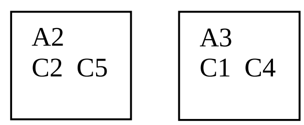


Goal State

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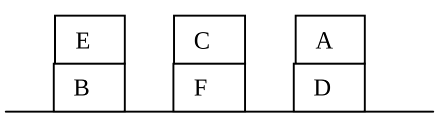
1. Explain the ways of improving the performance of the MINIMAX algorithm. (3)
2. How game playing exhibits features of search problems? (2)
3. What are the components of an action schema? (1)
4. Generate an optimal plan through Forward State Space search using PDDL for the planning problem in the Blocks world presented below. Omit the predicate ‘Block’, and give priority to gain in actions over gain in time. (4)

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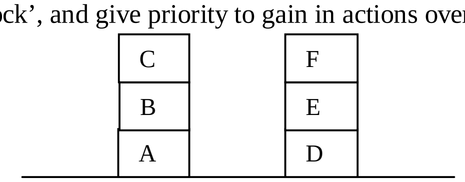


Goal State

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1. Explain how actions are carried out to get to the next state during planning. (3)
2. What strategies for generating optimal plans do you know? (2)
3. What is a cooperative game? (1)
4. Construct a pruned game tree using Alpha-Beta pruning. Take the sequence, [5, 2, 3, 4, 1, 3, 6, 2, 8, 4, 5, 7, 3, 1] of MINIMAX values for the nodes at the cutoff depth of 4 plies. Assume that branching factor is 2, MIN makes the first move, and nodes are generated from *right to left.* (4)

1. Explain the advantages and disadvantages of MINIMAX search strategy. (3)
2. What makes the search space complicated in game playing? (2)
3. What is a planner? (1)
4. Generate an optimal plan using PDDL and Forward State Space search for the Air Cargo Transportation problem given below. Omit predicates ‘Airport’, ‘Cargo’ and ‘Plane’, and consider generally accepted rules and notations for such a problem. (4)


Initial State Goal State

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1. Explain how MINIMAX values are assigned to nodes of game trees. (2)
2. How time constraint affects game playing? (2)
3. Why a planner is called a goal based agent? (2)
4. Generate an optimal plan through Forward State Space search using PDDL for the planning problem in the Blocks world presented below. Omit the predicate ‘Block’, and give priority to gain in number of actins over time. (4)


Initial State Goal State

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1. Explain the concept of action schemas using an example from Tower Building in Blocks World problem. (2)
2. How a planner exerts intelligence through domain feature consideration? (2)
3. What is quiescence search? (2)
4. Construct a pruned game tree using Alpha-Beta pruning. Take the sequence, [6, 3, 2, 6, 4, 5, 7, 2, 8, 7, 5, 1, 3, 4] of MINIMAX values for the nodes at the cutoff depth of 4 plies. Assume that branching factor is 2, MAX makes the first move, and nodes are generated from *left to right.* (4)

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1. Explain two important aspects of MINIMAX search algorithm. (2)
2. Why game playing is considered a complicated search problem? (2)
3. How ‘precondition’ of actions are represented with PDDL? (2)
4. Generate an optimal plan using PDDL and Forward State Space search for the Air Cargo Transportation problem given below. Omit predicates ‘Airport’, ‘Cargo’ and ‘Plane’, and consider generally accepted rules and notations for such a problem. (4)


Initial State Goal State

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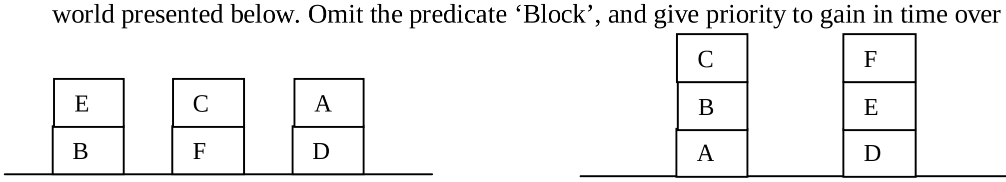
1. Explain the common factors considered for improving the performance of the MINIMAX algorithm. (2)
2. How game playing may be both cooperative and competitive? (2)
3. What is the purpose of action schemas? (2)
4. Generate an optimal plan through Forward State Space search using PDDL for the planning problem in the Blocks world presented below. Omit the predicate ‘Block’, and give priority to gain in time over number of actions. (4)




Initial State Goal State

1. Explain how ‘effect’ of an action is represented and used in planning. (2)
2. What makes a planner intelligent? (2)
3. Why players of a two-player board game are called MAX and MIN? (2)
4. Construct a pruned game tree using Alpha-Beta pruning. Take the sequence, [5, 3, 2, 6, 4, 5, 7, 1, 8, 7, 5, 1, 3, 4] of MINIMAX values for the nodes at the cutoff depth of 4 plies. Assume that branching factor is 2, MAX makes the first move, and nodes are generated from *right to left*. (4)

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1. Explain the advantages of MINIMAX search strategy. (2)
2. What do you know about multi-agent environment in game playing? (2)
3. How goal states are represented in planning problems? (2)
4. Generate an optimal plan using PDDL and Forward State Space search for the Air Cargo Transportation problem given below. Omit predicates ‘Airport’, ‘Cargo’ and ‘Plane’, and consider generally accepted rules and notations for such a problem. (4)


Initial State

Goal State

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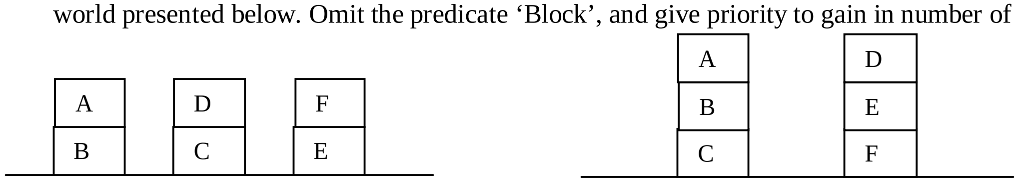
1. What are the disadvantages of MINIMAX strategy and how those can be reduced? (4)
2. How ‘precondition’ makes a planner intelligent? (2)
3. Generate an optimal plan through Forward State Space search using PDDL for the planning problem in the Blocks world presented below. Omit the predicate ‘Block’, and give priority to gain in number of actins over time. (4)




Initial State

Goal State

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1. What is an action schema and how is it used to carry out an action? (4)
2. Elaborate the concept of quiescence search. (2)
3. Construct a pruned game tree using Alpha-Beta pruning. Take the sequence, [8, 5, 2, 6, 7, 5, 4, 1, 8, 7, 5, 1, 3, 4] of MINIMAX values for the nodes at the cutoff depth of 4 plies. Assume that branching factor is 2, MAX makes the first move, and nodes are generated from *right to left*. (4)

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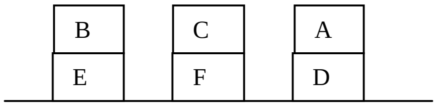
1. How game playing is converted to search problem and what peculiarities does such a search possess? (4)
2. Distinguish between total-order and partial-order plans. (2)
3. Generate an optimal plan using PDDL and Forward State Space search for the Air Cargo Transportation problem given below. Omit predicates ‘Airport’, ‘Cargo’ and ‘Plane’, and consider generally accepted rules and notations for such a problem. (4)

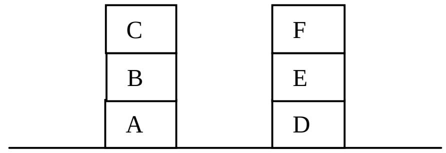

Initial State

Goal State

1. What is alpha-beta pruning and how a pruned game tree is constructed? (4)
2. Elaborate the concept of imperfect information considered in game playing. (2)
3. Generate an optimal plan through Forward State Space search using PDDL for the planning problem in the Blocks world presented below. Omit the predicate ‘Block’, and give priority to gain in time over number of actions. (4)



Initial State



Goal State

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1. How states and actions are represented using PDDL? (4)
2. Write a short note on ‘Decomposition in designing a plan’. (2)
3. Construct a pruned game tree using Alpha-Beta pruning. Take the sequence, [1, 3, 4, 5, 7, 4, 6, 2, 1, 3, 5, 4, 8, 4, 6] of MINIMAX values for the nodes at the cutoff depth of 4 plies. Assume that branching factor is 2, MIN makes the first move, and nodes are generated from *left to right.* (4)

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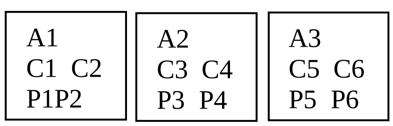
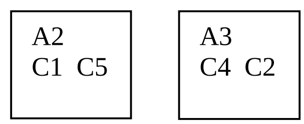
1. Illustrate with an example how a two player game can be represented and strategies of the players formulated. (4)
2. How do you explain the closed world assumption? (2)
3. Generate an optimal plan using PDDL and Forward State Space search for the Air Cargo Transportation problem given below. Omit predicates ‘Airport’, ‘Cargo’ and ‘Plane’, and consider generally accepted rules and notations for such a problem. (4)



Initial State



Goal State